Foundamentals of life insurance pricing

Calculating the actuarial present value (APV)

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Introduction

- **Premium**: Price of the insurance. Is the amount of money that the policyholder pays so that, in exchange, the insurer will pay in case that the claim takes place.

$$Premium = Pure\_premium + Management\_expenses + Tax + others$$

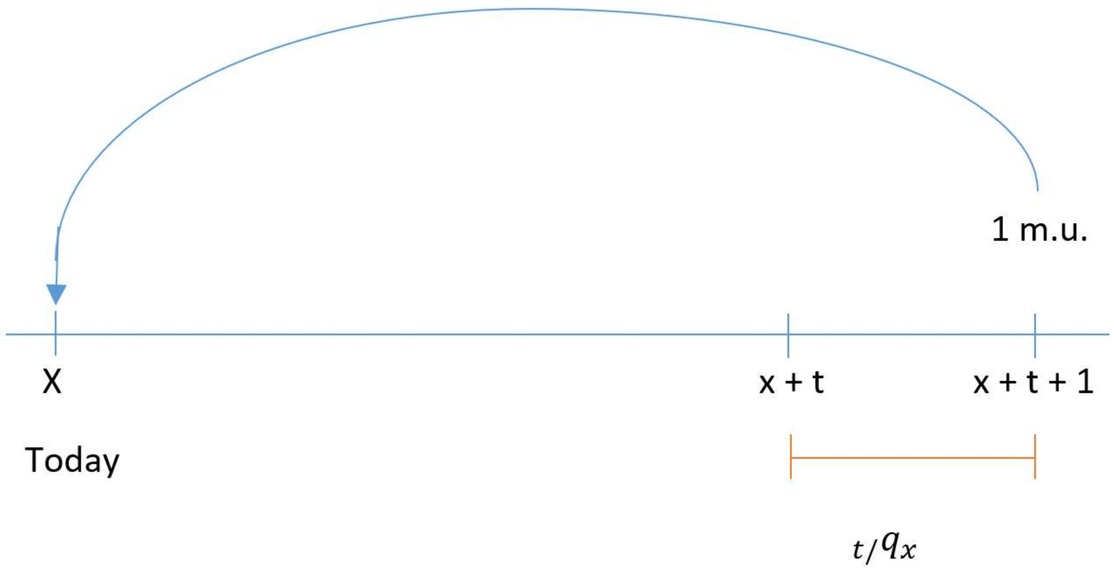
- **Pure premium**: It comes from the concept of mathematical expectation or expected value of the real cost that the insurer assumes.
- **Actuarial statistics**: Branch of statistics that deals with the determination of the probabilities of the occurrence of the events which are covered by the policy.
- **Life actuarial statistics**: it focusses on life insurance contracts
- **Non life actuarial statistics**: it focusses on the rest of policies, such as the automobile or household insurance.

Introduction

- **Insurance pricing (ratemaking)**: Is the determination of the rates, or premiums, that the insurer charge for the insurance policies. The difference with respect to other products is that the actual cost of providing the coverage is unknown (until the policy period has elapsed). For that reason, insurance rates are based on predictions rather than actual costs. Two types of assessments are required in that process:
- **Stochastic assessment**: We have to calculate the probability that the event (death) will occur during the coverage period.
- **Financial assessment**: The contract is underwritten a specific day (for example today), but the benefits are paid to the beneficiaries at some moment in the future. For that reason, it is necessary to calculate which is the present value of the future benefits.
- Based on these two assessments, we calculate the so-called **actuarial present value (APV)** of a contract.

APV calculation

- **Benefit payable at the end of the year of death**: Life insurance benefit of 1 monetary unit (m.u.) payable at $t+1$ if the insured aged x dies between age $x+t$ and $x+t+1$. The APV of the benefit can be calculated as follows, where I_1 is the annual interest rate:

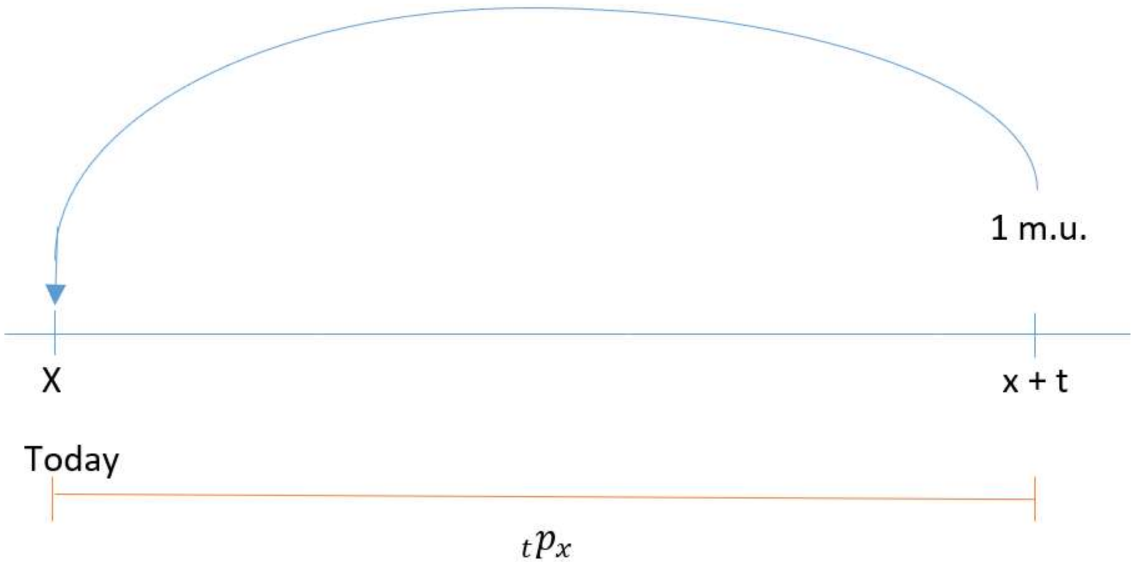


Present values	Probabilities
0	$1 - {}_{t/}q_x$
$(1 + I_1)^{-(t+1)}$	$_{t/}q_x$
	Sum = 1

$$\text{APV} = {}_{t/}q_x \cdot (1 + I_1)^{-(t+1)}$$

APV calculation

- **Pure endowment**: It is an endowment payable at the end of the policy period if the insured is alive. If the insured has died, there is no payment. If we assume that the payment is 1 m.u. payable at $x+t$ if the insured survives, the APV of the benefit can be calculated as follows:

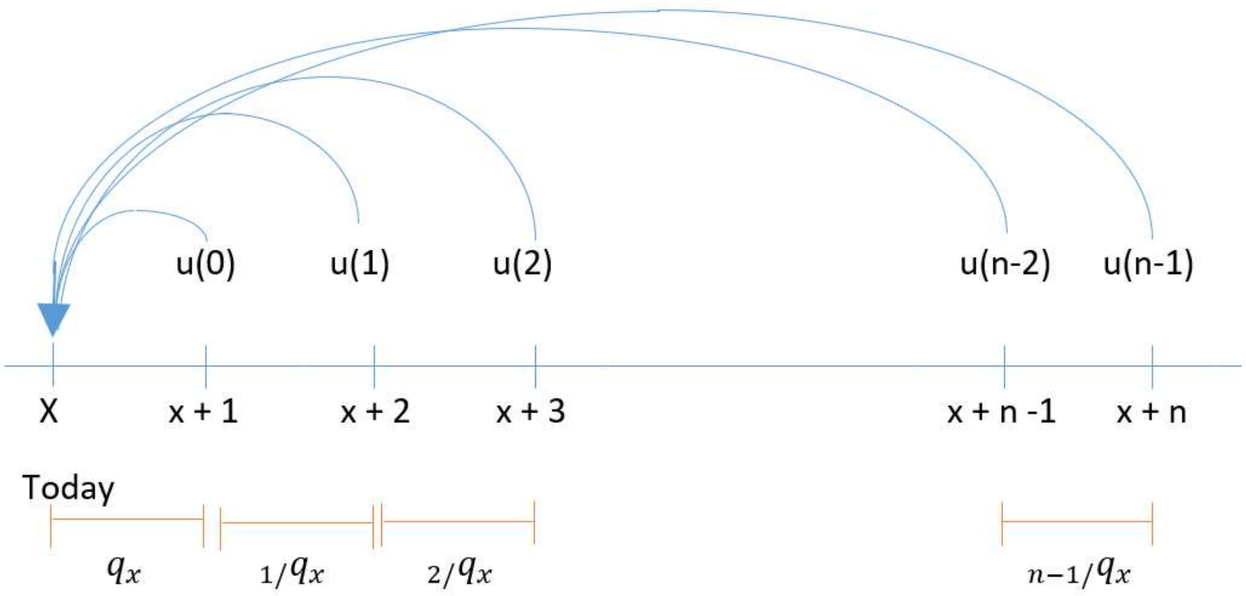


Present values	Probabilities
0	$1 - {}_t\mathrm{p}_x$
$(1+I_1)^{-t}$	${}_t\mathrm{p}_x$
	Sum = 1

$$\mathrm{APV} = {}_t\mathrm{p}_x \cdot (1+I_1)^{-t}$$

APV calculation

- **One-life insurance**: Life insurance benefit of $u(t)$, $t = 0,..,n\text{-}1$ monetary units (m.u.) payable at the end of the death year, for an insured aged x if he dies in next n years. The APV of the benefit can be calculated as follows, where I_1 is the annual interest rate:



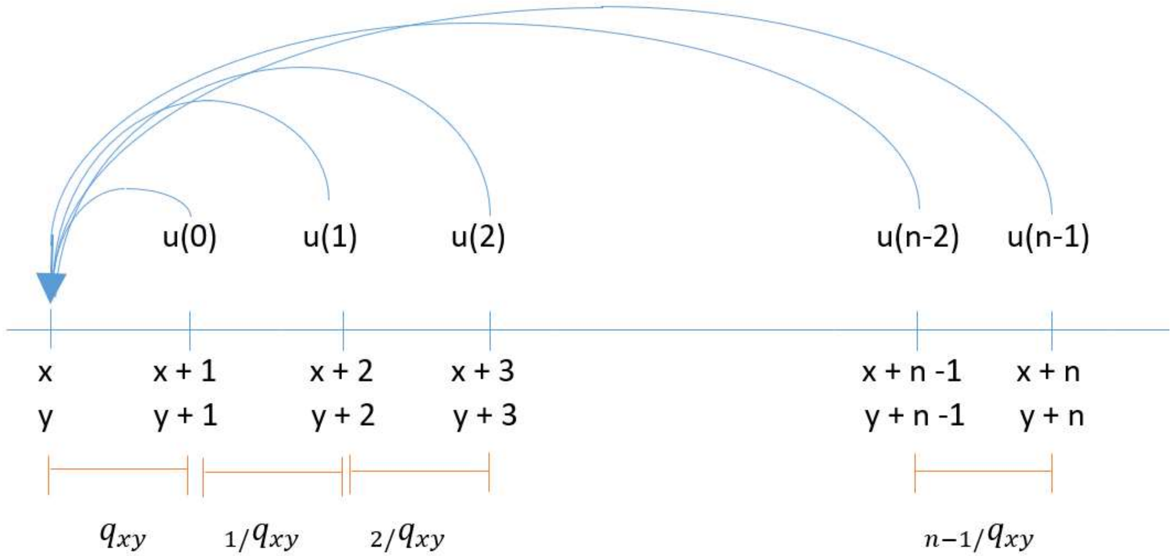
APV calculation

Present values	Probabilities
$u(t)(1+I_1)^{-(t+1)}$	${}_{t/}q_x \quad t = 0, \dots, n-1$
0	${}_{\mathrm{n}}\mathrm{p}_x$
	Sum = 1

$$\mathrm{APV} = \sum_{t=0}^{n-1} {}_{t/}q_x \cdot \mathrm{u(t)} \cdot (1+I_1)^{-(t+1)}$$

APV calculation

- **Two-lifes insurance**: Life insurance benefit of $u(t)$, $t = 0,..,n\text{-}1$ monetary units (m.u.) payable at the end of the dissolution (or extinction) year, for an couple aged (x, y) if the dissolution (or extinction) occurs in next n years. In case of dissolution, the APV of the benefit can be calculated as follows, where I_1 is the annual interest rate:



APV calculation

Present values	Probabilities
$u(t)(1+I_1)^{-(t+1)}$	${}_{t/}q_{xy} \quad t = 0, \dots, n-1$
0	${}_{\mathrm{n}}\mathrm{p}_{xy}$
	Sum = 1

$$\mathrm{APV} = \sum_{t=0}^{n-1} {}_{t/}q_{xy} \cdot \mathrm{u(t)} \cdot (1 + I_1)^{-(t+1)}$$

Example

For an insured aged 56, calculate the following probabilities:

- Probability of death before age 58
- Probability of death between 58 and 61

Calculate the APV of the following contract, that this insured aged 56 is going to underwrite:

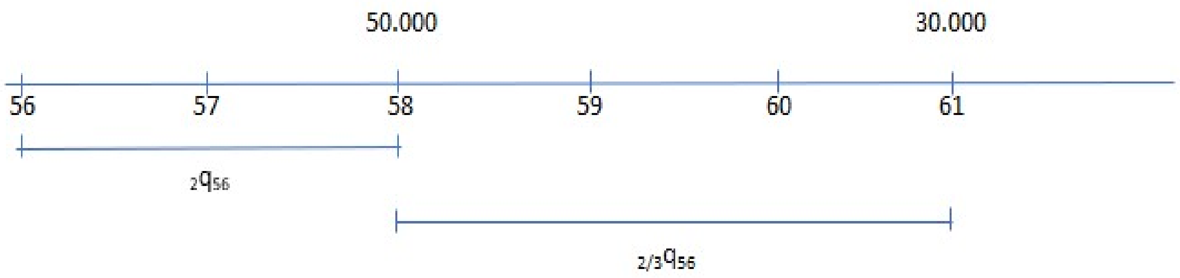
- If he dies before 58, a benefit of 50000 euros will be payd when the insured would have reached 58
- If he dies between 58 and 61, a benefit of 30000 euros will be payd when the insured would have reached 61

Which is the probability that the Insurance company will not pay any benefit?

The interest rate = 2%, the life table is..

Age (x)	q_x
55	0,0104
56	0,0112
57	0,0117
58	0,0121
59	0,0124
60	0,0129
61	0,0132
62	0,0136

Example



Present values	Probabilities	Present values · Probabilities
$50000(1{,}02)^{-2}$	$_2q_{56}$	$50000(1{,}02)^{-2} \cdot {}_2q_{56}$
$30000(1{,}02)^{-5}$	$_{2/3}q_{56}$	$30000(1{,}02)^{-5} \cdot {}_{2/3}q_{56}$

$_2q_{56} = 1 - {}_2p_{56} = 1 - p_{56} \cdot p_{57} = 1 - (1 - q_{56}) \cdot (1 - q_{57}) = 1 - 0{,}977231 = 0{,}022769$

$_{2/3}q_{56} = {}_2p_{56} \cdot {}_3q_{58} = 0{,}977231 \cdot (1 - {}_3p_{58}) = 0{,}977231 \cdot (1 - p_{58} \cdot p_{59} \cdot p_{60}) = 0{,}036095$

P(no payment) = 1- 0,022769 - 0,036095 = 0,941136

APV = 50000 · $1{,}02^{-2}$ · 0,022769 + 30000 · $1{,}02^{-5}$ · 0,036095 = 2075,007 euros